

NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Seventh Semester B.E. (Credit System) Degree Examinations

January 2023

19BT8X05 – NANOTECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Compare bottom-up and top-down approach.	10	L*2	1	1
	b) Schematically explain the process of nanowire synthesis.	10	L2	2	1
2.	a) Explain the nanosphere lithography techniques.	10	L2	1	1
	b) Write a note on				
	(i) Fullerenes				
	(ii) Thin films	10	L2	2	1
3.	a) With suitable example explain the optical property of nanomaterial	10	L2	1	1
	b) Explain the principle and working of atomic force microscopy	10	L2	2	1
Unit – II					
4.	a) What is microfluidics? List and explain the different microvalves	10	L2	3	1,9
	b) Explain the two surgical applications of MEMS	10	L2	4	1,9
5.	a) List out the microvalve available for microfluidics devices. Explain its working.	10	L2	3	1,9
	b) Write a note on				
	(i) Electromagnetic transducer				
	(ii) Mechanical transducer	10	L2	4	1,9
6.	a) Schematically explain the fabrication of microfluidic channel using polymer.	10	L2	3	1,9
	b) Explain two recent developments of MEMS.	10	L2	4	1,9
Unit – III					
7.	a) Explain the application of nanomaterials in apparel industry.	10	L2	5	1,12
	b) Write a nanomaterial application note on				
	(i) Data storage				
	(ii) Cosmetics	10	L2	5	1,12
8.	a) Discuss the ethical aspects involved in nanotechnology.	10	L2	5	1,12
	b) Illustrate the application of nanotechnology in the food and agriculture sector.	10	L2	5	1,12

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
